Recursive Partitioning Approaches to Modeling Genetics as a Source of Treatment Effect Heterogeneity

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"Now, the world don't move to the beat of just one drum,

What might be right for you, may not be right for some."

-- 'It Takes Diff'rent Strokes,' by Alan Thicke,

Gloria Loring and Al Burton)

Introduction:

Treatment effects may be heterogeneous for a variety of reasons -- some idiosyncratic and some systematic. The field of pharmacogenomics has recently begun to identify genetic predictors of systematic variation in treatment outcomes (Evans and McLeod, 2003, Evans and Relling, 1999). In the broader field of statistical genetics, markers of genetic heterogeneity measured across the entire human genome are being interrogated for associations with likelihood of developing disease (Duerr et al., 2006, Hirschhorn and Daly, 2005, Lin et al., 2004, Sladek et al., 2007). Other biomarkers, including those indicative of oxidative stress (Montuschi et al., 2000, Nielsen et al., 1997) and broader biological frailty (such as allostatic load) (Johnson, 2006, Seeman et al., 2001, Seplaki et al., 2004) are also shown to underlie heterogeneous health outcomes.

Notably, many of these pathways are indirect: the genes and biomarkers identified themselves may not act causally, but instead they may serve as proxies for direct causes through potentially complex networks (Alm and Arkin, 2003). Furthermore, gene-gene, gene-environment and higher-order interactions of genes and biomarkers have been shown to predict both response to treatment and development of disease, even though they may not in fact be causal (Coffey et al., 2004, Hahn et al., 2003, Kraemer, 2001). These complex non-causal relationships pose challenges to the standard statistical, epidemiologic and econometric models upon which cost-effectiveness analysis (or indeed, any form of comparative analysis) is based.

In this paper, we outline a two-step approach that identifies possibly complex and high-order interactions of dichotomous covariates that are predictive of heterogeneous treatment effects (HTE) and incorporates the resulting classification information into regression-based cost-effectiveness analysis. The first step uses an ensemble of recursive partitions with random variable selection, with some similarities to Breiman's Random Forest classifier (Breiman, 2001), while the second step uses Bayesian generalized linear modeling (GLM) to predict the posterior probability of treatment cost-effectiveness conditional on classifying information derived from the first step. We use Monte Carlo simulated genetic data to investigate how our our new recursive partitioning ensemble approach might perform (compared to the Random Forest classifier and stepwise logistic regression) in the context of cost-effectiveness analysis conducted alongside a clinical trial where pharmacogenomic data are available.

II. Recursive Partitioning

Recursive partitioning (RP) (Zhang and Singer, 1999) comprises a set of non-parametric methods for partitioning groups of individuals, j = 1 to N, into subgroups with similar outcomes, Y, based on a set of covariates X_k, $k = 1$ *to* K. The most well-known recursive partitioning method, Classification and Regression Trees (CART), was developed by Leo Breiman and colleagues in the early 1980s (Breiman et al., 1984).

In RP, a parent node τ_P (group of individuals) is split into two child nodes (subgroups) based on the values of a covariate X_k. Individuals with values of X_k less than some cutoff value c then belong to the left child node, $\tau_L = \{j \mid j \in \tau_P, X_{j,k} < c\}$ while those with $X_k \geq c$ belong to the right child node, $\tau_R = \{j \mid j \in \tau_P, X_{j,k} \geq c\}$. Each child node either becomes an internal parent node that is then further split or becomes a terminal node. The recursive nature of the splitting process produces a bifurcated "tree" whose terminal nodes partition the original dataset.

In "greedy" RP algorithms, at each parent node τ_P, X_k and c are chosen to maximize the reduction of heterogeneity (or "impurity") of Y in the child nodes relative to the parent node. In the case of dichotomous outcomes Y, let $p = \text{Prob}(Y = 1 \mid \tau) = (1/N_\tau) \Sigma_{j=1:N} Y_j$, where j indexes the N_τ individuals in node τ. Then impurity of a node, $i(\tau) = \varphi(p)$, where $\varphi(p) \geq 0$, $\varphi(p) = \varphi(1-p)$ and $\varphi(0) = \varphi(1) < \varphi(p)$. Commonly used measures of impurity include entropy, $\varphi(p) = [-p \log(p)]-[(1-p)\log(1-p)]$ and the Gini index, $\varphi(p) = p(1-p)$. For other categorical or continuous outcomes, alternative measures of impurity (e.g., information entropy, sum of squared deviations, etc.) are frequently used (Zhang and Singer 1999).

The reduction in impurity induced by a split on variable X_k at cutpoint c is

(1) $$\Delta i(k,c,\tau) = i(\tau) - [(N_{\tau L}/N)i(\tau_L) + (N_{\tau R}/N)i(\tau_R)]$$

For categorical or continuous covariates, generally an optimal c_k is chosen first for each covariate X_k; then the covariate that maximizes $\Delta i(k,c_k,\tau)$ is selected. In greedy algorithms, a node will not be partitioned if $\Delta i \leq 0$ for all k and c_k. Note that in the case of dichotomous covariates X (which will be the focus of our Monte Carlo example), $c = 1$ by default.

Any resulting child node can in turn become a parent node (and hence be further partitioned) unless either outcomes Y or covariates X are completely homogeneous. However, fully-grown trees may be difficult to interpret and run the risk of over-fitting the data.

The CART method balances model fit with tree parsimony and stability by "cost-complexity pruning" fully-grown trees. For a tree, T, define total tree cost-complexity

(2) $$R_\alpha(T) = \sum_{\tau \in \widetilde{T}} p(\tau) r(\tau) - \alpha \left|\widetilde{T}\right|$$

where $\widetilde{T}$ is the set of terminal nodes of T, $p(\tau)$ is the fraction of total observations belonging to node τ, α is a complexity penalty, $|\widetilde{T}|$ is the number of terminal nodes in T and $r(\tau) = (1/N_\tau)\Sigma_{j=1:N_\tau}\{\rho(\hat{Y}_j - Y_j)\}$ measures the average cost of misclassification (or incorrect prediction) of individuals in node τ.

In the case of dichotomous outcomes or classes (which is the focus of this paper), it is common practice to assign the predicted outcome/class $\hat{Y}_j = 1$ if the median (or some other pre-designated percentile if one observed outcome is relatively uncommon) outcome for the terminal node to which j belongs equals 1, and $\hat{Y}_j = 0$, otherwise. Class probabilities can be estimated by assigning $\hat{Y}_j = (1/N_\tau)\Sigma_{j'=1:N\tau}\{Y_{j'}\}$. Misclassification costs $\rho(\hat{Y}_j - Y_j)$ may then be calculated for each individual, with $\rho(-1) > 0$ being the cost of a false negative, $\rho(1) > 0$ being the cost of a false positive and $\rho(0) = 0$ for a correctly-classified case. For class probabilities or other continuous outcomes, squared or absolute prediction error are commonly used as cost functions.

The fully-grown tree T may be pruned by converting any internal parent node into a terminal node (hence deleting any child nodes as well as their successors, if any). Breiman and colleagues (1984) demonstrated that for any value of the complexity penalty α, there is a unique smallest subtree of T, T_α* that minimizes cost-complexity.

An alternative method of tree pruning based on statistical measures of dissimilarity between child nodes was first contemplated by Segal (1988) in the context of RP-based survival analysis and later extended by Zhang and Singer (1999) to classification trees for dichotomous outcomes. This method was successfully used by Zhang and Bonney (2000) to identify two disease-causing alleles in a genome-wide scan of the Genetic Analysis Workshop 9 (GAW-9) simulated dataset.

After constructing a fully-grown tree T, a raw goodness-of-split score S_τ is assigned to each internal parent node τ_P equal to the χ^2 statistic for the 2 x 2 contingency table formed by its child nodes τ_L and τ_R. In the simple example in table 1, $S_{\tau P}$ = ((10-20)^2)/20 + ((40-30)^2)/30 + ((20-10)^2)/10 + (5-15)^2/5 = 37.35

[Table 1 About Here]

Then, we calculate a maximum score for each τ_P

(3) $$S_{\tau P}^{\max} = \max_{\tau \in \Psi} \{S_\tau\}$$

where Ψ consists of all parent nodes of the proper sub-tree rooted at τ_P. The maximum score reflects that a split is judged no worse than the best split produced by its offspring. Any parent node whose maximum score is less than some selected χ^2 critical value (say, corresponding to $p < 0.05$ in a one-sided test) is pruned (including any subsequent splits). This pruning method allows for poor splits to occur early in the tree, so long as they lead to good splits later on. Application of the maximum score algorithm is shown in figure 2 from Zhang and Bonney (2000) reproduced below as figure 1.

[Figure 1 About Here]

Choice of pruning parameters influences overall size of the final optimal sub-tree (with smaller α or χ^2 critical values yielding larger final trees). Because estimates of the misclassification cost based on the same data used to build the tree (referred to as resubstitution cost) are biased downward, pruning parameters selected by this method tend to produce trees that are larger than optimal (Zhang and Singer, 1999).

To address this concern, pruning parameters are often chosen through cross-validation (CV), generally proceeding as follows. The original dataset is partitioned into M datasets, indexed by $m = 1$ to M. First, datasets $m = 1$ through $M-1$ are combined into a learning dataset used to generate predicted outcomes at each node $\tau \in T$ (i.e., the full tree that was constructed using the entire dataset). Then, for each of a sequence of pruning parameters, the optimal sub-tree is determined, thus deriving a sequence of optimal subtrees of T, each corresponding to a particular pruning parameter parameter. For each optimal sub-tree, observations from test dataset M are then allocated into the terminal nodes based on values of their covariates X, and a non-penalized misclassification cost $R_0(T)$ is calculated. The whole procedure then repeats with $m = 2$ *to* M combined as the learning set and $m = 1$ as the test set; etc. Finally, misclassification cost corresponding to each pruning parameter in the sequence is averaged over the M CV repetitions, and the parameter with the lowest average misclassification cost is selected.

III. Ensemble methods.

While the CV methods used to select pruning parameters can overcome the underestimation of true misclassification cost, they do not address concerns about the structure of the fully-grown tree T, itself. Spurious relationships in the full dataset may lead to a fully-grown tree that includes unrelated covariates (or excludes covariates that are actually related to the outcome).

Berk (2006) suggests the logic that if averaging resubstitution costs constructed from different, non-overlapping subsets of data can correct for overly optimistic estimates of true misclassification cost, perhaps averaging over a set of full trees (each constructed from different data) might correct for overly optimistic trees. Dietterich (2000) defines ensemble methods to be "learning algorithms that construct a set of classifiers and then classify new data points by taking a (weighted) vote of their predictions."

An early and still popular ensemble approach applicable to RP is the method of "bagging" (Bootstrap AGgregatING) developed by (Breiman, 1996). When bagging trees, the original dataset is first split into a learning dataset $\mathcal{L}$ and a test dataset $\mathcal{T}$. $\mathcal{L}$ is then randomly sampled with replacement, and the resulting bootstrap dataset, $\mathcal{L}_B$ is used to construct a full tree T_B using impurity-based methods. The original learning dataset $\mathcal{L}$ is then used as a test set to prune T_B (by misclassification cost, χ^2 goodness-of-split, etc.). This process is repeated $B = 2$ *to* N_B to yield a sequence of optimal pruned tree classifiers, $\{T_B*\}$. For each individual $j \in \mathcal{T}$, predicted class or outcome is usually assigned based on the median class (or class probability, or mean outcome) for the terminal node $\tau_\omega \in T_B*$ to which individual j belongs, for each tree in the sequence $\{T_B*\}$. Then, an overall class or outcome prediction is constructed by taking the median predicted class or mean predicted class probability (or other continuous outcome) over the entire sequence of optimal subtrees.

Bagging and other forms of model aggregation based on resampling a training dataset have been shown to consistently stabilize prediction by equalizing the influence of observations in the training set (Grandvalet, 2004). Breiman (1996) notes that with respect to classification, assigning class probabilities rather than assigning predicted class based on majority or plurality yields similar results. However, testing performance is more difficult because while true class may be observed for each member of the test dataset, true class probability is unobservable.

IV. Random Forests.

Ensembles built on CART (and similar greedy RP tree-building algorithms) may still have difficulty classifying individuals when covariates are only associated with outcomes through higher-order interactions. Consider the following simple example. Suppose that individuals are partitioned into two subset "classes," C_1 and C_2 and that:

$$(4) \qquad p(Y_j = 1 \mid j \in C_1) = p1 > p(Y_j = 1 \mid j \in C_2) = p2$$

That is, class C_1 individuals have a higher probability of $Y = 1$ than class C_2 individuals). Also suppose that for $j \in C_1$, $X \sim N_K[0,\Sigma_1]$ and $j \in C_2$, $X \sim N[0,\Sigma_2]$, $\Sigma_1 \neq \Sigma_2$, where $N_K[\mu,\Sigma]$ is a fairly large (say, $K = 10$) dimensional, normally-distributed row-vector with mean μ and covariance matrix Σ. Note that by construction, $Y \perp X_k$ for all k. Given the differing covariance structures of X between high and low response types, some higher-order interaction could potentially produce a useful split. However, greedy algorithms would be unexpected to find a single covariate X_k that produces an improvement in the impurity score for the root node. In practice, whether or not the useful higher-order interaction is detected would seem to depend on pure luck.

Indeed, recent developments in classification algorithms have relied on randomization not only of included observations, but also of included covariates (random feature selection), to overcome this concern (Dietterich, 2000). As with bagging, the Random Forest classifier (Breiman, 2001) uses bootstrap resamples of a learning dataset to construct full-sized trees (Breiman et al., 1984). Rather than using a greedy algorithm that selects the best possible cutpoint and predictor, Random Forests use a different random subset of the available covariates at each parent node. Breiman (2001) notes that the classification error in Random Forests decreases in the strength of the individual trees in the forest and increases in the correlation of their predictions. Trees may be strengthened by pruning or by selecting larger random subsets of covariates, but at the expense of a higher degree of correlation in the predictions (which reduces the benefit of bagging).

Finding an appropriate balance between tree strength and correlation of predictions may be a problem-driven exercise. Fortunately, the Random Forests algorithm allows for estimates of both tree strength and prediction correlation to be obtained during the course of estimation from "out-of-bag" observations in the learning set $\mathcal{L}$ that are not selected into $\mathcal{L}_B$ (the bootstrapped sample from $\mathcal{L}$, with replacement) (Breiman, 2001).

However, a limitation to all ensemble methods is that it is very difficult (if not impossible) to interpret the classifying relationship between X and Y (Berk, 2006, Dıá, 2005). A single, modest-sized classification tree is easily interpretable – a classification dependent on the pooled "votes" of possibly hundreds or thousands of such trees cannot be described simply in terms of X.

V. Applications to Pharmacogenomics

We will be applying a Random-Forest-like RP algorithm to a simulated pharmacogenomic data set to determine whether the method has the potential to identify indirect associations between genes and treatment response. Pharmacogenomics is the study of inherited differences in the response to drugs (Evans and McLeod 2003). It makes use of high throughput genetic sequencing of the entire human genome to identify genetic variation associated with heterogeneous treatment effects. The primary targets of pharmacogenomics are single nucleotide polymorphisms or SNPs (for description, see figure 2).

[Figure 2 About Here]

The earliest findings in pharmacogenomics involved variations in single genes known to encode specific drug-metabolizing enzymes (Evans and McLeod, 2003, Evans and Relling, 1999, Roden et al., 2006). However, the promise of personalized drug treatment based on the causal "candidate-gene" approach has not been realized, in part due to previously underappreciated complexity of the metabolic pathway, interactions of genes with the environment and other yet undiscovered factors (Clayton et al., 2006).

New, "genome-wide association" approaches recognize that interactions of genes (including those whose function is not yet well understood) are likely to either directly cause treatment effect heterogeneity, or at the very least carry information about unknown causal factors (Pearson and Manolio, 2008, Roden et al., 2006). Due to way that genes are physically located on a chromosome and passed on to offspring in sexual reproduction, SNPs in the human genome are highly correlated. Variants that are inherited together are called "haplotypes" (Goldstein and Cavalleri 2005). Because of the high degree of physically-induced correlation, only a relatively small number of characteristic ("tag") SNPs is sufficient to capture most of haplotype structure of the human genome (Zhang et al., 2004).

VI. Simulated Pharmacogenomic Dataset

The International HapMap Project identifies tagSNPs across the entire human genome in individuals from four populations from different parts of the world and freely releases that information into the public domain through the World Wide Web (International HapMap Project, 2008). Using the January 2007 release of HapMap, we selected two 100Kb (Kb = 1,000 base-pairs) regions on 120 copies (pertaining to 60 individuals, since each individual has a pair of chromosomes) of the 158.6Mb (million base-pair) Chromosome 7. These regions were selected because they had very high levels of heterogeneity, measured by the density of SNPs per 500Kb and also were associated with known genes.

In each region, we obtained phased haplotype data for all identified tagSNPs (14 in region A: 89637569 to 89729569 and 20 in region B: 126359539 to 126455333). Although it is not relevant to our simulation, it is perhaps interesting to note that Region A (see figure 3) contains the claudin 12 gene (CLDN12), which has been shown to play a role in colorectal cancer (Grone et al., 2007), while Region B (see figure 4) is located within the glutamate receptor 8 gene (GRM8), which plays a role in normal brain chemistry has shown variation among individuals at risk of developing heroin addiction (Nielsen et al., 2008), autism (Muhle et al., 2004) and other neuropsychiatric disorders.

[Figures 3 and 4 About Here]

We used the HapGen program (Marchini, 2007) to simulate 400 individual genotypes (each genotype consists of two phased haplotypes – one for each allele) containing the 34 above-discussed tag SNPs, conditional on the 120 observed haplotypes from the HapMap project. HapGen uses a hidden Markov model approximation to the population genetics model of Li and Stephens (2003) to simulate the effect of meiotic recombination of DNA that would occur over successive generations of sexual reproduction.

We then selected one of the tagSNPs (locus ch7: 126408607) to be associated with individual likelihood of response to treatment 1, and another tag SNP (locus ch7: 89687508) to be associated with individual likelihood of response to treatment 2. For each locus, an individual's genotype can either be homozygous wild-type (both alleles are normal), heterozygous (one allele is normal and one has a polymorphism) or homozygous variant (both alleles have a polymorphism). We assume that for both treatment response loci, homozygous wild-types, heterozygous and homozygous variants have a 0%, 50% and 100% chance, respectively, of being a "high responder" phenotype. Note that an individual can be a high responder phenotype to one treatment and normal responder phenotype to the other. Being a high responder phenotype does not guarantee a response to treatment. Normal responders have a 20% and 30% chance of response to treatments 1 and 2, respectively, while high responders have 3-fold higher probability of response (60% and 90% to treatments 1 and 2, respectively). See Table 2 for distribution of genotypes, phenotypes and treatment responses in the simulated dataset.

[Table 2 About Here]

Because our goal is to test the performance of our methods in the context of cost-effectiveness analysis, we additionally simulate costs and quality-adjusted life years (QALYs) associated with response or non-response under treatments 1 and 2.

Costs (divided by 10,000) are simulated using gamma distributions, C~Gamma$[r,\theta_{T,R}]$ with shape parameter, $r = 5$, and scale parameters $\theta_{1,0} = 5$, $\theta_{2,0} = 2.5$, $\theta_{1,1} = 10$, $\theta_{2,1} = 3.3$, where for each $\theta_{Tr,R}$, $Tr \in \{1,2\}$ denotes treatment and $R \in \{0,1\}$ denotes actual treatment response. Scale parameters are set such that for individuals who did not respond, average costs are twice as much on treatment 2 as on treatment 1. Relative to non-responders, average cost for individuals who responded to treatment 1 is reduced by half, while average cost for those who responded to treatment 2 is reduced by 25%.

QALYs are simulated using beta distributions, Q ~ Beta$[n\alpha_R ,n(1-\alpha_R)]$ with $n = 50$, $\alpha_0 = 0.1$, $\alpha_1 = 0.6$. Average QALYs are 0.1 and 0.6 for individuals who responded and did not respond, respectively, and otherwise do not vary by treatment.

The parameters of the simulation (including Hapgen simulated genotypes) are set such that, on average, treatment 2 is net-beneficial (relative to treatment 1) for the general population only if a decision-maker is willing to pay over ~$140,000 per QALY (see Figure 5). However, this threshold is lower for individuals who have a high response phenotype to treatment 2 (~60,000 if they are also high response phenotype to treatment 1; ~$20,000 if they are a normal phenotype to treatment 1). Treatment 2 is never net beneficial for individuals who are high response phenotype to treatment 1, but normal phenotype to treatment 2.

[Figure 5 About Here]

VII. Methods: RP algorithm for genetic scoring (Step One)

The RP algorithm we use in this specific context of genetically-determined subgroups for cost-effectiveness analysis predicts class probability, $P(Z_{j,Tr} = 1|X_j)$, which is the probability that an individual with a vector X_j of dichotomous tag SNP variables is likely to have higher than average response rates to treatment *Tr*. The RP algorithm of this paper varies from Random Forests in two fundamental ways.

First, rather than choosing a random subset of predictor variables at each interior node (random feature selection) and then choosing the best splitting variable among that subset, this algorithm simply chooses a splitting variable at random from the entire set of predictor variables without regard to how well the variable performs. Technically speaking, this still fits within the definition of random feature selection, with a singleton subset – but practically speaking, the Random Forest classifier typically uses a larger subset of predictor variables. Larger subsets tend to produce individual trees that are stronger classifiers, but produce ensembles of trees whose predictions are more highly correlated. At the upper limit where the subset of covariates is the entire set, the method of Random Forests reduces to bagging. As discussed above, the out-of-sample performance of ensemble methods increases with the strength of each individual classifier, but decreases with stronger correlation across predictions. Our choice of complete random tree structure minimizes correlation across predictions, but at the expense of individual tree performance.

Our second variation from Random Forests is that we attempt to strengthen the performance of each tree through pruning while the Random Forests approach uses fully-grown trees in order to reduce correlation of predictions across trees. In essence, our approach to balancing classifier strength and correlation of predictions (random feature selection from singleton sets; pruned trees) is the reverse of the approach taken by Random Forests (random feature selection from larger sets; un-pruned trees).

Our class probability algorithm proceeds as follows:

1. Split original dataset into a learning dataset $\mathcal{L}$ and a training dataset $\mathcal{T}$ of equal size.

2. Generate a sequence $\{T_t\}$, $t = 1$ to T_{max} (= 500,000) of completely random, fully-grown trees with fixed recursion depth λ (= 4). Note that a tree with recursion depth λ consists of a sequence $\{\tau_p\}$ of $2^\lambda-1$ internal nodes and a sequence $\{\tau_\omega\}$ of 2^λ terminal nodes. Therefore, each node in each tree we generate can represent up to a 4-way interaction of SNPs. At each τ_p we select splitting covariate SNP X_k where k belongs to the subset of all covariates that were not used to generate a split of any parent of τ_p and the probability of choosing each k in the subset is equal. Note that if our covariates were continuous or categorical rather than dichotomous, this restriction would be relaxed. We would also need to choose split points c for each potential splitting covariate – a process that could be conducted based on measures of impurity, perhaps with a random disturbance.

3. Prune each full tree $T \in \{T_t\}$ into pruned tree T^P using the χ^2 goodness-of-split method of Zhang and Singer (1999), with critical value $\chi^2_{a,1}$ (where a is the

size of a one-sided χ^2 test and 1 denotes the number of degrees of freedom). In our example, we choose $a = 0.99$, which yields a critical value of 6.635. We also impose a size restriction such that parent nodes containing fewer than 15 observations will be pruned. By convention, we assign a score of 0 to any pruned parent nodes.

4. Collect all pruned trees T^P and eliminate any duplicate trees to create sequence $\{T'_t\}$.

5a. For each tree $T^P \in \{T'_t\}$, construct a sequence $\{\mathcal{L}_b\}$, $b = 1$ to B $(= 100)$ of bootstrap re-samples (with replacement) from learning dataset $\mathcal{L}$.

5b. For each $\mathcal{L}_b$, calculate a resubstitution score for the tree T^P using the sum of maximized split scores, with a penalty α for each non-pruned split:

$$R_\alpha(T^P \mid L_b) = \sum_{\tau p} \left(\mathrm{S}_{\tau p}^{\max} - \alpha \cdot 1[\mathrm{S}_{\tau p}^{\max} > 0] \right) \tag{5}$$

where $1[]$ is an indicator function. In our example, we chose $\alpha = \chi^2_{0.99,1}/2$ (one half the critical value used to prune parent nodes). Note that setting $\alpha < \chi^2_{0.99,1}$ allows a tree to offset the penalty for a poor split at τ_p by obtaining at least one acceptable split in any successor nodes that are subsequently split. All else equal, smaller α promotes trees for which higher-order interactions are necessary to produce good splits because lower-order interactions or direct associations do not. Such trees are, in general, larger and weaker classifiers, but they are also likely to produce less-correlated predictions than trees pruned with larger values of α.

5c. Calculate the mode of the kernel-smoothed density of resubstitution scores for tree T^P:

$$R_\alpha^*(T^P) = \left(\frac{1}{Bh}\right) \underset{R}{\arg\max} \left\{ \sum_{b=1}^{B} K\left(\frac{R - R_\alpha(T^P \mid L_b)}{h}\right) \right\} \quad (6)$$

where K is a standard normal kernel and h is a bandwidth selected according to Silverman's rule of thumb (Silverman, 1986).

6. Eliminate any trees with $R^*_\alpha(T^P) < 0 \; \chi^2_{0.99,1} =$ 6.635 to obtain a "final committee" of trees $\{T^*_t\}$.

7a. For each tree $T^P \in \{T^*_t\}$, construct a sequence $\{\mathcal{L}_c\}$, $c = 1$ to C $(= 50)$ of bootstrap re-samples (with replacement) from learning dataset $\mathcal{L}$.

7b. For each individual $k \in \mathcal{T}$ (the test dataset), let $\tau_{T,k}$ denote the terminal node in tree T to which individual k belongs. For each replicated learning dataset $\mathcal{L}_c$, calculate the overall proportion of responders, μ_c, and the proportion of responders in node $\tau_{T,k}$:

(7) $$\mu_{\tau:k,c} = \left(\frac{1}{N_{\tau:k,c}}\right)\sum_{m=1}^{N_{\tau:k,c}}\{Y_m\}$$

where m indexes the $N_{\tau:k,c}$ individuals from replicated learning dataset $\mathcal{L}_c$ that belong to $\tau_{T,k}$. Assign predicted class:

(8) $$\hat{Z}_{\tau:k,c} = 1 \text{ if } \mu_{\tau:k,c} > \mu_c \text{ and } \hat{Z}_{\tau:k,c} = 1 \text{ if } \mu_{\tau:k,c} \le \mu_c$$

In other words, for each tree and each replication of the learning set, an individual in the test set is predicted to be a member of a high responder class if they fall in a terminal node with a greater than average proportion of individuals who responded.

7c. For each individual $k \in \mathcal{T}$, calculate the weighted average class probability score across trees and learning set replications, using tree resubstitution scores calculated in step 5c as weights:

(9) $$\hat{Z}_k = \left(\frac{1}{C}\right)\sum_{c=1}^{C}\left\{\frac{\sum_{T\in\{T^*\}}\left\{R_\alpha^*(T)\hat{Z}_{\tau:k,c}\right\}}{\sum_{T\in\{T^*\}}R_\alpha^*(T)}\right\}$$

VIII. Results: RP Algorithm (Step One)

We analyzed our simulated pharmacogenomic cost-effectiveness analysis dataset using the ensemble recursive partitioning approach outlined above, as well as the Random Forest classifier (package randomForest 4.5-23 for the R statistical system; Fortran original by Leo Breiman and Adele Cutler, R port by Andy Liaw and Matthew Wienerin) and stepwise logistic regression ('stepwise' in STATA 9.0).

We split our 400 simulated genotypes, phenotypes and outcomes into a learning and test dataset of $n = 200$, each. The dependent variable was an indicator ($R = 1$ if the individual responded to treatment, $= 0$ otherwise) and the predictor variables were the 32 tagSNPs from Regions A and B of Chromosome 7. Each SNP was recoded into two dichotomous variables (SNP_1 = 1 if one or two alleles are affected; SNP_2 = 1 if two alleles are affected; SNP_1 = SNP_2 = 0 if an individual is normal).

We also examined performance of each method in the case of a "negative control" in which Treatment 2 was simulated, but the underlying genotype (and subsequent phenotype and outcomes) was simulated from a completely random trinomial distribution. For the negative control case, observed genotype data should confer no classifying information.

For treatment 1, the 500,000 random trees were pruned into 66,207 non-degenerate trees, 4,628 of which were unique. After bootstrapping and selecting the final committee of trees (step 6), 954 unique trees remained. There were 895 unique 2-way SNP interactions and 594 unique 3-way interactions (figures 6 and 7) in the final committee. Each circle in figures 6 and 7 represents a SNP interaction that appears in one or more trees. The X-Y-(Z) coordinates represent the SNP (1-64) used as a primary, secondary (or tertiary) split. The size of the circle equals the sum of the modal resubstitution scores (step 5c) over all trees in which the interaction is found. The set of 2-way interactions suggests important classifying information is contained in SNPs 3, 20 and 45 (as primary splits), 22 and 45 (as secondary splits).

For treatment 2, the 500,000 random trees were pruned into 81,554 non-degenerate trees, 8,952 of which were unique. The final committee consisted of 3,201 unique trees, containing 1,693 unique 2-way interactions of SNPs (figure 8) and 1,205 unique 3-way interactions (figure 9). The set of 2-way interactions suggests important interactions between SNPs 18, 19, 27, 42-43 and 50 (as primary splits), and 34, 38, 43 and 50 (as secondary splits).

In the negative control case, the 500,000 random trees were pruned into 6,975 non-degenerate trees, 1,663 of which were unique. The final committee consisted of 100 unique trees, containing 184 unique 2-way interactions of SNPs (figure 10) and 110 unique 3-way interactions (figure 11). Even though there is no classifying information in the SNPs for the negative control, several interactions survive the pruning and scoring process (although there are many fewer than for simulated treatments 1 and 2). The set of 2-way interactions suggests the importance of SNPs 10 and 11 as primary splits and SNP 2 as a secondary split. Interestingly, the most significant interactions (between SNPs 41 and 43 and 43 and 52) appear to be sporadic, as these SNPs do not tend to appear in other recurring interactions.

We next examine the out-of-sample correlation between the predicted class probability score (or in the case of stepwise logistic, probability of response) and observed response in the test dataset (table 4). For both treatments 1 and 2, the score produced by the RP ensemble algorithm was more strongly correlated with out-of-sample outcomes than the scores produced by either the Random Forest classifier or stepwise logistic regression. Although we would not be able to observe the true latent genotype and phenotype in the real world, we can do so in our simulated dataset. We find that for both treatments 1 and 2, the scores from the RP ensemble algorithm are more strongly correlated with both the latent genotype and phenotype than both other methods.

Figure 12 presents the distribution of class probability scores under each algorithm for treatment 1, treatment 2 and control. We note that the distribution of scores under the RP algorithm is tri-modal (corresponding to the three underlying genotypes) for both treatments 1 and 2, but uni-modal for the negative control. The distribution of stepwise logistic scores is also tri-modal, but for treatment 1 only.

Figures 13-15 further decompose the scores by genotype, for each treatment. We note that the RP algorithm scores appear to be most cleanly separated by genotype for both treatments 1 and 2, but (correctly) not for the negative control. In particular, for treatment 2 (Figure 14), we see that no heterozygous individuals (with probability = 0.5 of being a high response phenotype) have a score lower than the highest homozygous wild-type (with probability = 0 of being a high response phenotype).

IX. Methods Bayesian GLM for cost-effectiveness (Step Two)

Finally, we use Markov Chain Monte Carlo (MCMC) to estimate generalized linear models (GLM) of treatment response, cost and QALYs in a Bayesian framework. These estimates were conducted out-of-sample, using the test dataset (*n = 200*). The use of Bayesian methods allows us to make direct posterior probability statements about functions of parameters, including cost-effectiveness of treatment 2 relative to treatment 1 conditional on genetic scores. We generally follow approaches outlined by O'Hagan et al. (2001) and Vanness and Kim (2002).

Specifically, we model response using the Bernoulli family and probit link function:

$$R_i \sim \text{Bernoulli}\left[P_i\right] \tag{10}$$

$$P_i = \Phi(\beta_0 + \beta_1 Tr_i + \beta_2 (1 - Tr_i)\, Score0_i + \beta_3\, Tr_i\, Score1_i$$

where Φ is the standard normal distribution function, $Score0_i$ and $Score1_i$ are the class probability scores for individual *i* under treatment 1 and treatment 2, respectively, calculated using the RP algorithm, Random Forest classifier or stepwise logistic regression. The response model is estimated 4 times, once for each of the 3 scoring methods being compared, and a fourth time using the actual (ordinarily unobservable) individual response phenotypes under each treatment in place of *Score0* and *Score1*.

We model cost with the Gamma family and log link function:

$$(11) \qquad C_i \sim \text{Gamma}\left[(1-Tr_i)\,r_0 + Tr_i\,r_1,\,(1-Tr_i)\,\theta 0_i + Tr_i\,\theta 1_i\right]$$

$$\ln(r_0/\theta 0_i) = \gamma 0_0 + \gamma 0_1 R_i$$

$$\ln(r_1/\theta 1_i) = \gamma 1_0 + \gamma 1_1 R_i$$

Finally, we model QALYs with the Beta family and logit link function:

$$(12) \qquad Q_i \sim \text{Beta}\left[(1-Tr_i)\,a0_i + Tr_i\,a1_i,\,(1-Tr_i)\,b0 + Tr_i\,b1\right]$$

$$\text{Logit}\left(a0_i/(a0_i+b0)\right) = \delta 0_0 + \delta 0_1 R_i$$

$$\text{Logit}\left(a1_i/(a1_i+b1)\right) = \delta 1_0 + \delta 1_1 R_i$$

We assigned non-informative priors to our unknown parameters: N[0,1000] for β, γ, *and* δ and Uniform[0,1000] for b and r. When non-informative priors are used, Bayes' Formula implies that the posterior distribution is approximately proportional to the likelihood function – hence, our results should be comparable to GLM estimated by maximum likelihood. Of course, an advantage to the Bayesian approach is the ability to incorporate prior information from previous studies (say, through meta-analysis).

We used WinBUGS 1.4.3 both to estimate our GLM specification and to perform simulated treatment assignment to individuals with the highest (1) and lowest (0) possible genetic scores for each treatment. Because these simulations are conducted using draws from the parameter posterior distributions, they reflect second-order (parametric) uncertainty and allow us to make statements about the probability of cost-effectiveness for these subgroups of individuals. However, it is important to note that the method does not account for the sampling variability of the genetic scores themselves.

X. Results (Step Two)

Convergence of the single Markov Chain sampler was assessed visually by examining the history of draws; convergence appeared to be obtained after approximately 500 draws. To be conservative, we discarded the first 6,000 draws as a "burn-in" and conducted parameter inference and simulation using an additional 10,000 draws. Monte Carlo error was approximately 5% of the parameter standard deviation, or less. Parameter estimates and credible intervals can be found in table 4.

In general, the posterior 95% credible intervals for the association of higher class probability scores with probability of response under treatments 1 and 2 lie entirely above zero for all 3 methods. The credible intervals cross zero for all 3 methods under the negative control. The model using RP ensemble scores appears to most reliably obtain parameter estimates comparable to the model that used actual (ordinarily unobservable) class – particularly the estimate for the overall effect of treatment 2 relative to treatment 1 and the association of class probability on response to treatment 1.

Posterior expected net benefit curves (with 95% credible intervals) for treatment 2 relative to treatment 1 are presented in figure 16 (for individuals most likely to be high responders to treatment 1 and normal responders to treatment 2) and figure 17 (for individuals most likely to be normal responders to treatment 1 and high responders to treatment 2). All three genetic scoring methods uncovered the basic relationship built into the simulation (recall figure 5). The RP algorithm scores appear to have resulted in an over-estimate of the negative net benefit associated with choosing treatment 2 for individuals who are likely to be high responders to treatment 1 and normal responders for treatment 2. However, the RP algorithm scores also provided the closest estimate (and tightest posterior credible intervals) of the positive net benefit associated with choosing treatment 2 for individuals who are likely to be high responders to treatment 2 and normal responders to treatment 1. In particular, only the RP algorithm approach is able to make the claim of 95% or higher posterior probability that the net benefit of treatment 2 is positive, given the commonly-used willingness-to-pay threshold of $50,000 per QALY.

While results are promising, it is not known to what extent the small differences in method performance are due to sample variation. Further study on the statistical properties of the scores themselves is warranted.

XI. Discussion and Conclusion

Classical, experimental design is well-suited to obtaining unbiased estimates of an average treatment effect (ATE), even when outcomes vary systematically by observable or unobservable variables. However, patients are not likely to consider themselves merely average, and it is often said that "no doctor treats the average patient." If the direction and magnitude of systematic variation in treatment effects could be identified and demonstrated to have predictive validity, it would undoubtedly be of value to clinical decision-makers (Kravitz et al., 2004).

The existence of volumes of genome data are calling into question traditional hypothesis-testing driven research (Goodman, 1999). The potential even exists in the near term for genome data to become part of regular clinical practice. A

$500,000 prize has been offered by the J. Craig Venter Science Foundation for the first team that can offer individual whole-genome sequencing costing less than $1,000 (Service, 2006).

However, given the complexity of relationships underlying HTE discussed above, it is unlikely that the correct model can be specified before, say, a clinical trial takes place. Unfortunately, the classical hypothesis testing paradigm frowns upon *ex post* searches for HTE for reasons both practical and philosophical. Philosophically, once an experiment is complete, the data collected is no longer random; therefore, no probability statements can be made about whether the data at hand conform to particular statements about unknown parameters of interest. Such probability statements can only be made *ex ante*.

Practically, *ex post* specification searches are dangerous because of the increased chance of committing at least one Type-I error when testing for differences in treatment effects between subgroups implied by the final model specification. Bonferroni's correction is designed to preserve an overall acceptable rate α for committing at least one such error (Bland and Altman, 1995).

Unfortunately, Bonferroni's correction also increases the probability of committing a Type-II error. In general, classical hypothesis testing (and, in particular, Bonferroni's correction) elevates the status of the null hypothesis in a manner that may make inference unhelpful (Perneger, 1998). After all, a decision-maker choosing among a set of alternative treatments to maximize expected utility would not likely be concerned about an arbitrary threshold of confidence, above which one choice emerges as clearly superior to another. Such a decision-maker would simply like to choose the treatment with the highest expected utility. Health economist and member of the UK National Institute for Clinical Excellence Appraisal Committee, Karl Claxton has gone as far as to proclaim the "irrelevance of inference" in the real world of health technology assessment (Claxton, 1999).

Statistical decision theory based on Bayesian principles can potentially address both philosophical and practical concerns about *ex post* subgroup analysis. First, in Bayesian analysis, inference is made directly on the posterior probability distribution of the unknown parameters (or functions thereof). These parameters were random before the experiment (prior) and remain random afterwards (posterior); what changes with the collection of data is only the amount of uncertainty. Second, because we can test hypotheses about any function of parameters, we can directly assess posterior expected utility associated with each treatment option, conditional on any variable present in the model. The approach does not eliminate the increased potential of making a Type-I error; however, it does recognize that Type-II errors also have negative consequences.

Combining non-parametric classification methods with Bayesian models of quantities important for decision-making (like cost and QALYs associated with treatments) holds promise for better informing decision-makers who are coming to possess more and more pieces of the "genetic puzzle." Such decision-makers may not only be choosing which treatments to prescribe for their patients or cover for their policy-holders, but also may be designing the next clinical trial for a subgroup of interest (or deciding how much to pay for that trial). The approaches outlined in this paper will hopefully serve as a starting point for further evaluation and discussion.

Table 1.

	τ_L	τ_R
Y = 1	10	20
Y = 0	40	5

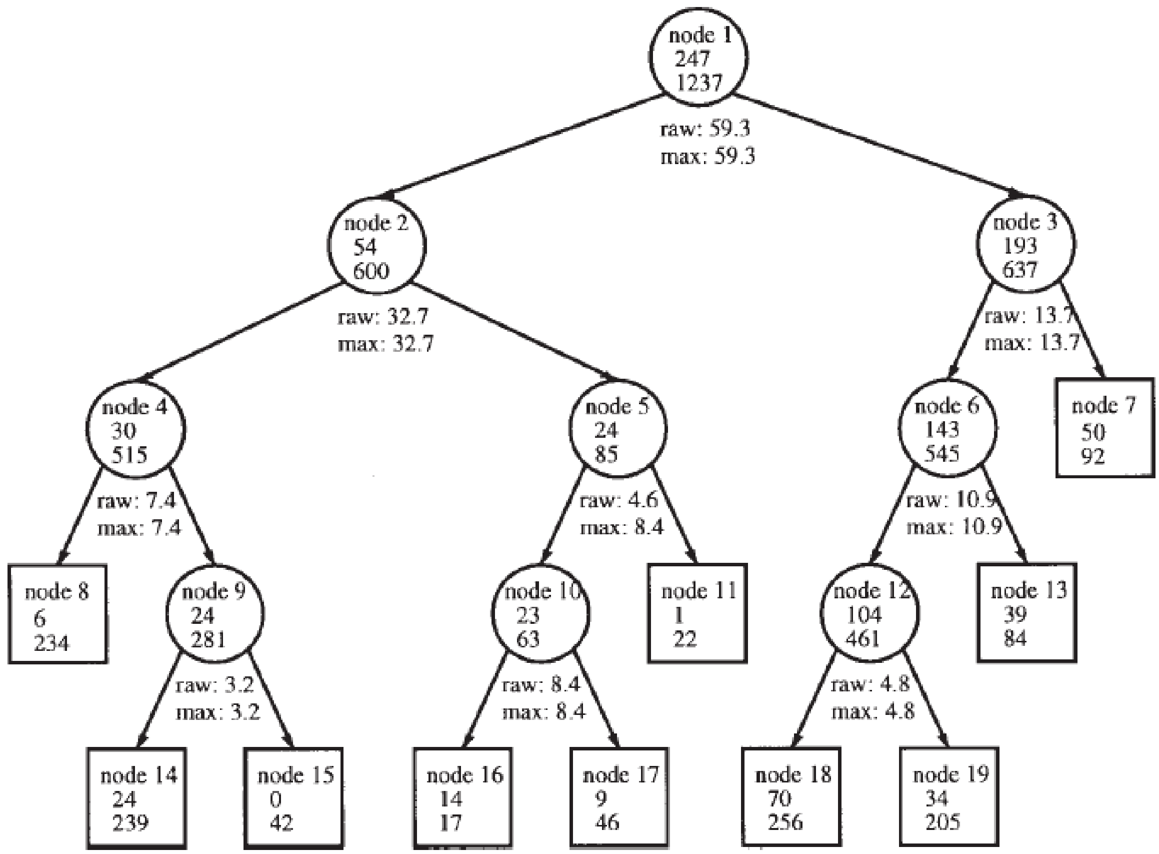


Fig. 2. Illustration of tree pruning. Inside each node are the node number **(top)**, the numbers of affected **(middle)**, and unaffected **(bottom)** individuals. Under each internal node is the raw and maximum χ^2 statistics, as described in the text in detail.

Figure 1. [MUST OBTAIN PERMISSION FROM PUBLISHER]

What are SNPs?
Single nucleotide polymorphisms or SNPs (pronounced "snips") are DNA sequence variations that occur when a single nucleotide (A,T,C,or G) in the genome sequence is altered. For example a SNP might change the DNA sequence A**A**GGCTAA to A**T**GGCTAA. For a variation to be considered a SNP, it must occur in at least 1% of the population. SNPs, which make up about 90% of all human genetic variation, occur every 100 to 300 bases along the 3-billion-base human genome. Two of every three SNPs involve the replacement of cytosine (C) with thymine (T). SNPs can occur in both coding (gene) and noncoding regions of the genome. Many SNPs have no effect on cell function, but scientists believe others could predispose people to disease or influence their response to a drug.

Source: Human Genome Project Information: http://www.ornl.gov/sci/techresources/Human_Genome/faq/snps.shtml#snps

Figure 2.

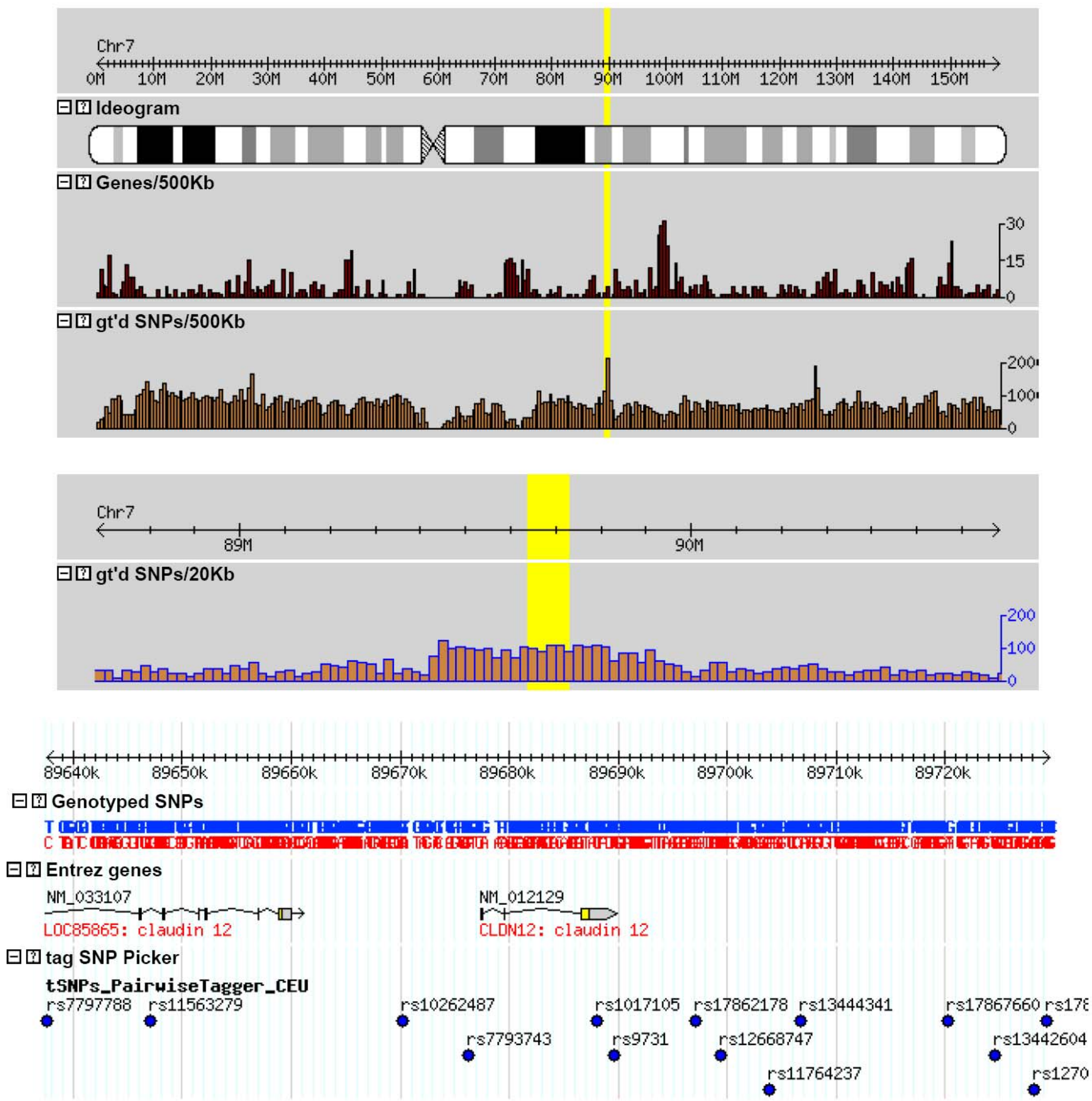


Figure 3. HapMap browser showing location of 14 tag SNPs in region A of Chromosome 7.

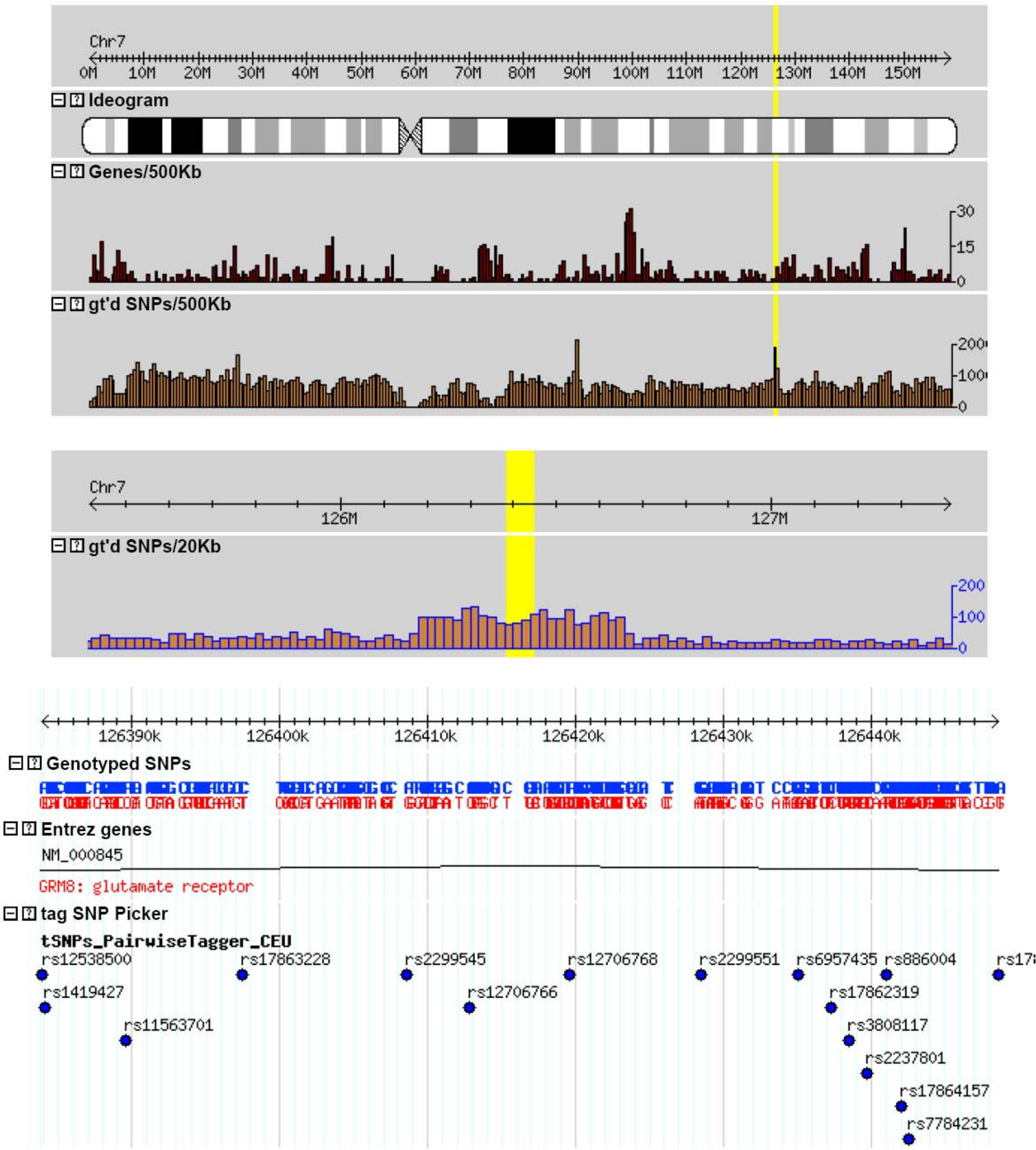


Figure 4. HapMap browser showing location of 20 tag SNPs in region B of Chromosome 7.

Table 2: Description of genotypes (homozygous wild-type, heterozygous, homozygous variant), phenotypes (high and normal responder) and outcomes (responded, did not respond) in learning and test datasets.

	Learning Dataset N	Test Dataset N	Learning Dataset % Responded	Test Dataset % Responded
Treatment 1	200	200	0.345	0.24
Homozygous Wild Type (P[HRP] = 0.0)	91	120	0.198	0.0833
Heterozygous (P[HRP] = 0.5)	94	69	0.447	0.406
Homozygous Variant (P[HRP] = 1.0)	15	11	0.6	0.909
High Responder Phenotype (Class 1)	67	51	0.687	0.667
Normal Phenotype (Class 0)	133	149	0.173	0.094
Responded	69	48	--	--
Did Not Respond	131	152	--	--
Treatment 2	200	200	0.515	0.345
Homozygous Wild Type (P[HRP] = 0.0)	107	105	0.336	0.2
Heterozygous (P[HRP] = 0.5)	81	81	0.716	0.444
Homozygous Variant (P[HRP] = 1.0)	12	14	0.75	0.857
High Responder Phenotype (Class 1)	51	49	0.902	0.776
Normal Phenotype (Class 0)	149	151	0.383	0.205
Responded	103	69	--	--
Did Not Respond	97	131	--	--
Treatment 2 (Control)	200	200	0.6	0.57
Homozygous Wild Type (P[HRP] = 0.0)	54	53	0.296	0.245
Heterozygous (P[HRP] = 0.5)	100	102	0.61	0.627
Homozygous Variant (P[HRP] = 1.0)	46	45	0.935	0.822

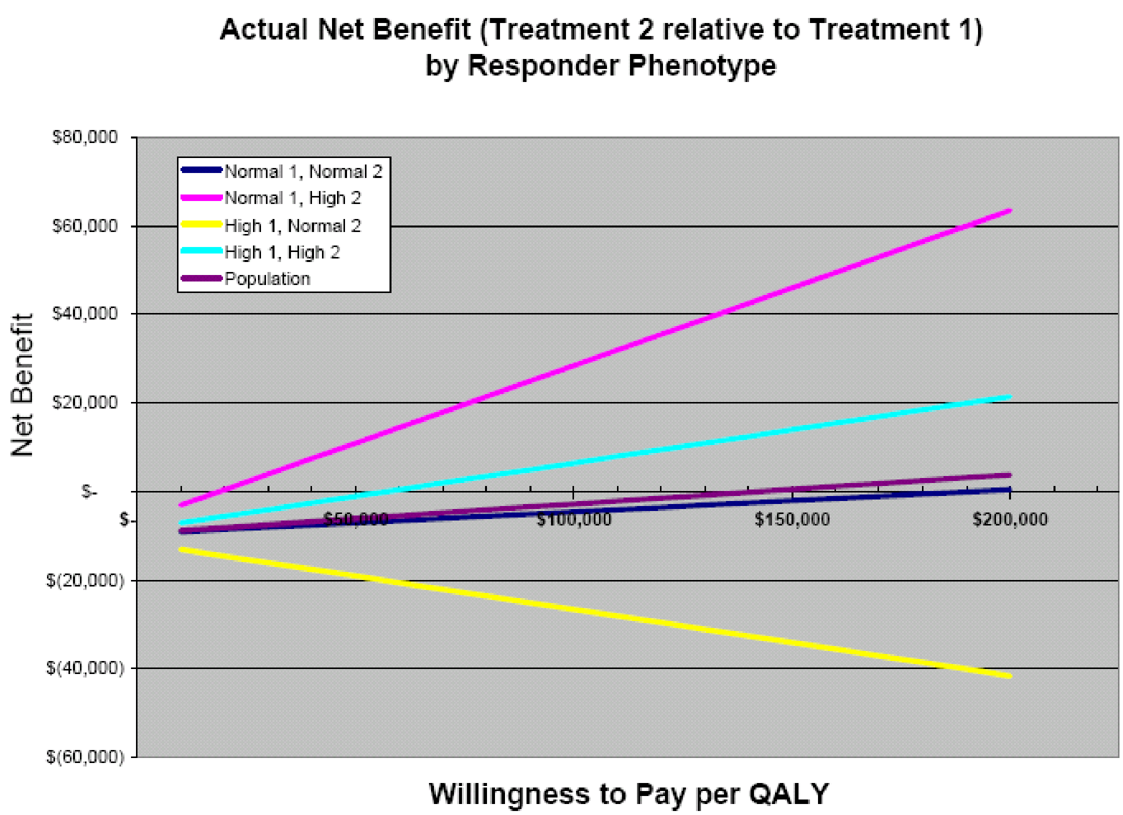
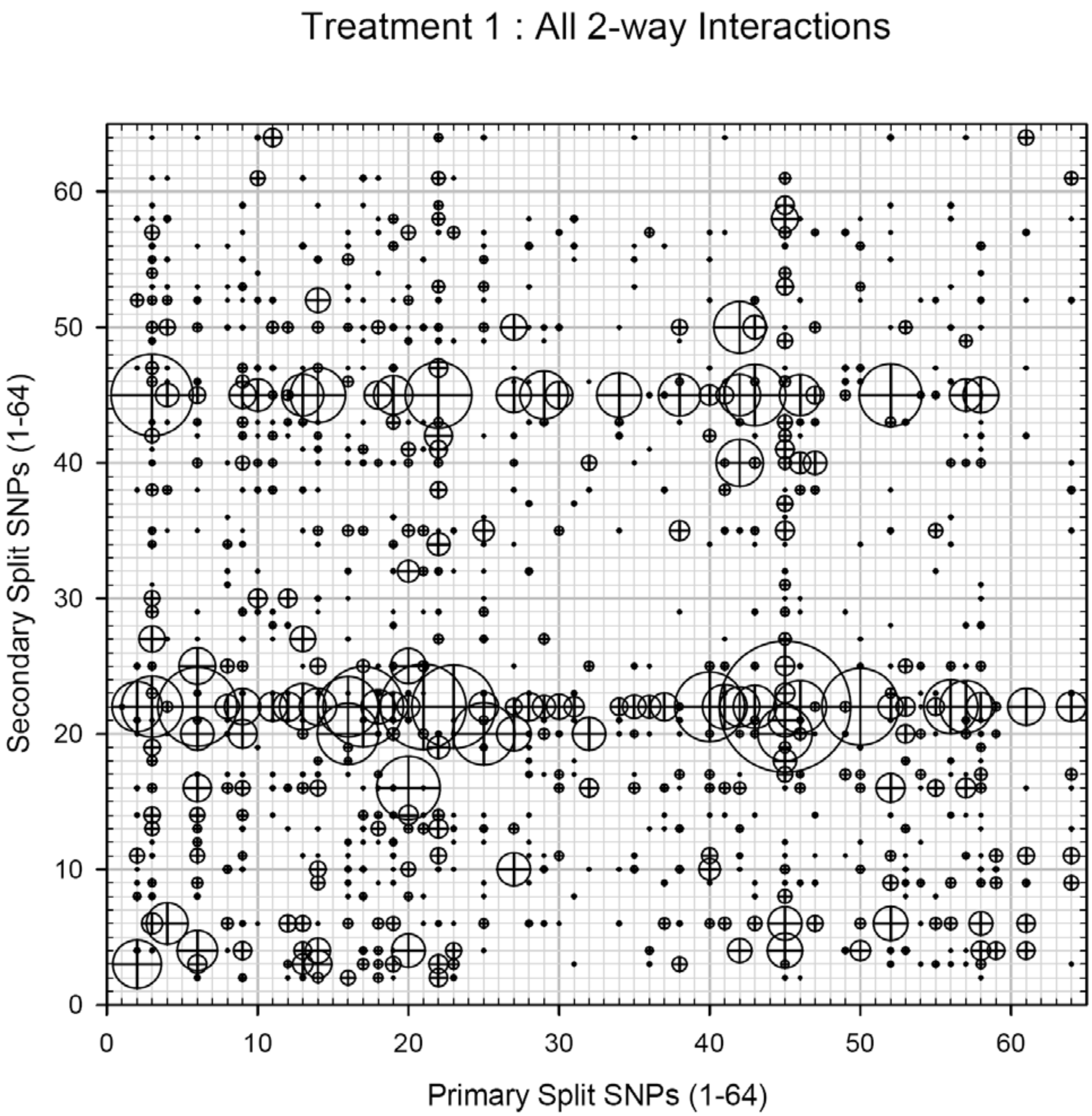


Figure 5. Actual expected net benefit curves, given simulation parameters.



Treatment 1 : All 2-way Interactions
Secondary Split SNPs (1-64)
Primary Split SNPs (1-64)
0
10
20
30
40
50
60

Figure 6.

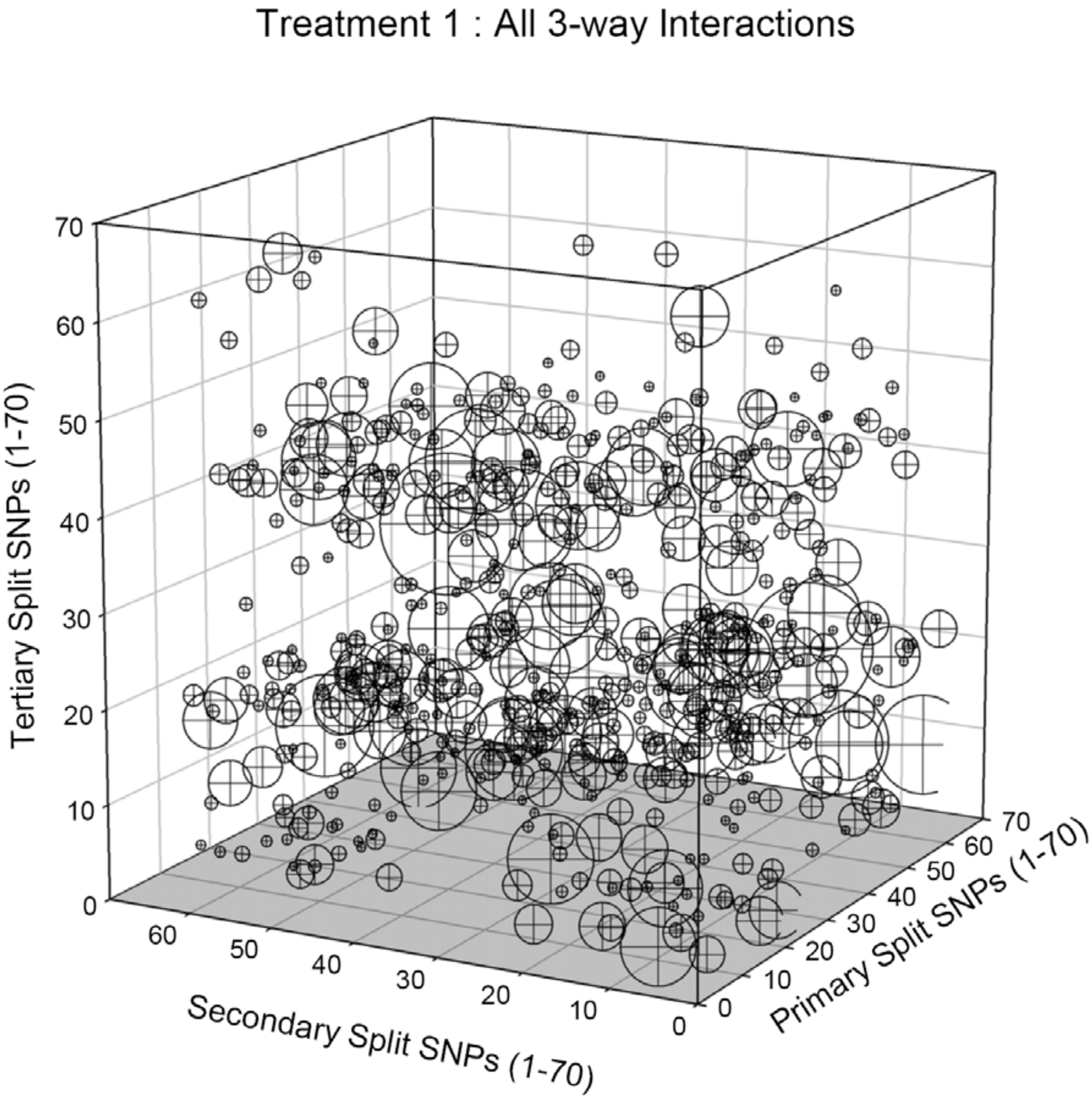
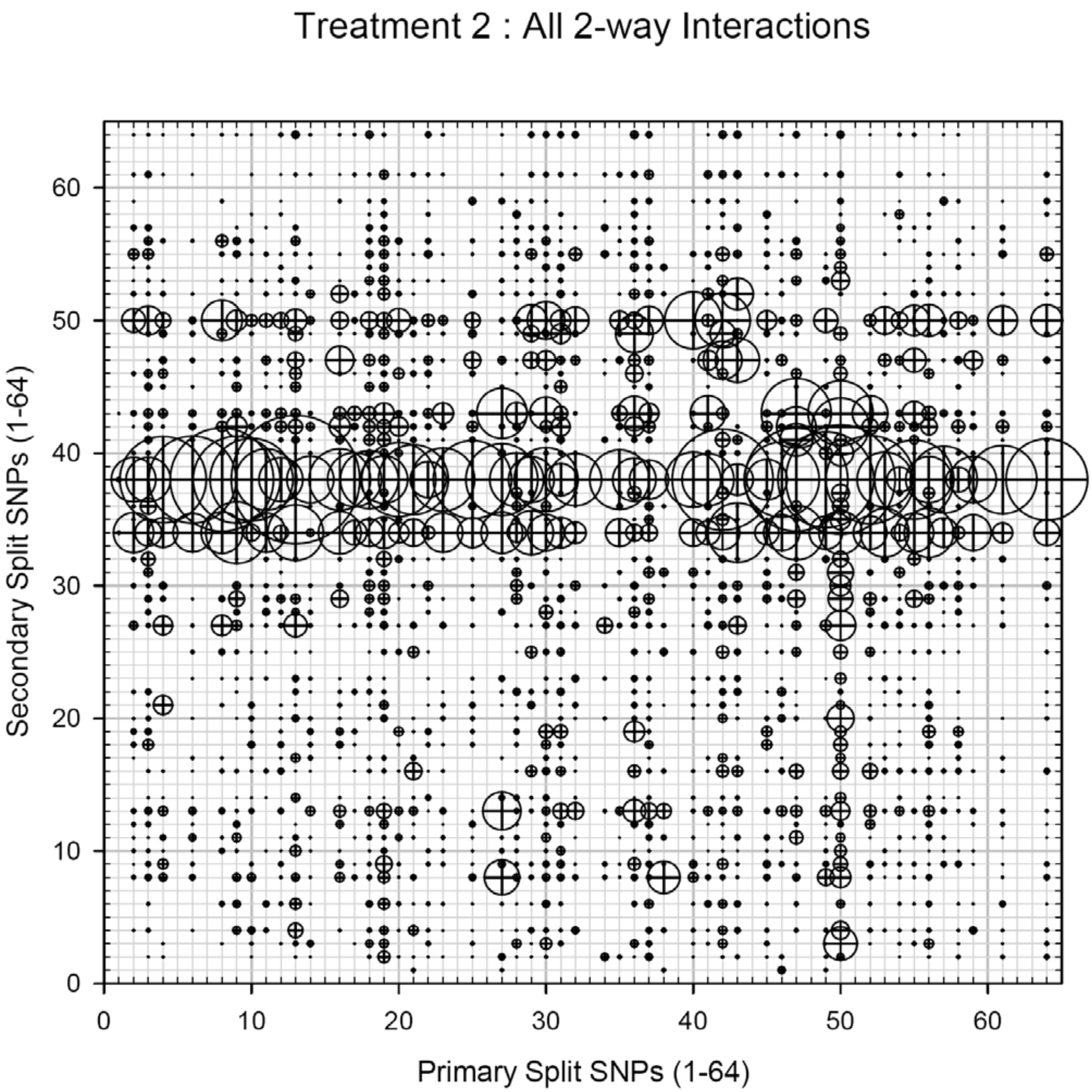

Treatment 1 : All 3-way Interactions
Tertiary Split SNPs (1-70)
70
60
50
40
30
20
10
0
60
50
40
30
20
10
0
Secondary Split SNPs (1-70)
70
60
50
40
30
20
10
0
Primary Split SNPs (1-70)

Figure 7.


Treatment 2 : All 2-way Interactions
Secondary Split SNPs (1-64)
Primary Split SNPs (1-64)
0
10
20
30
40
50
60
Figure 8.

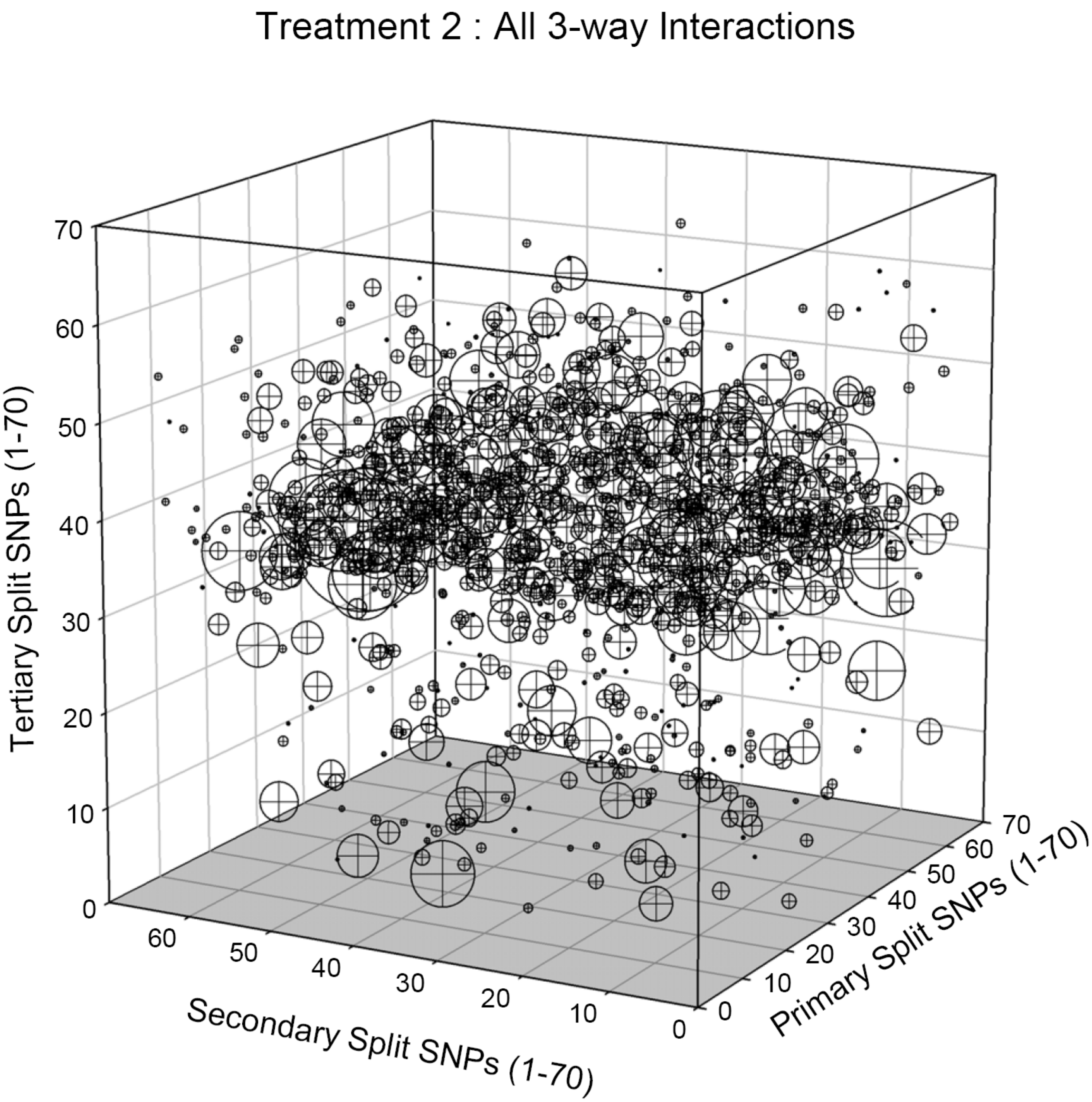
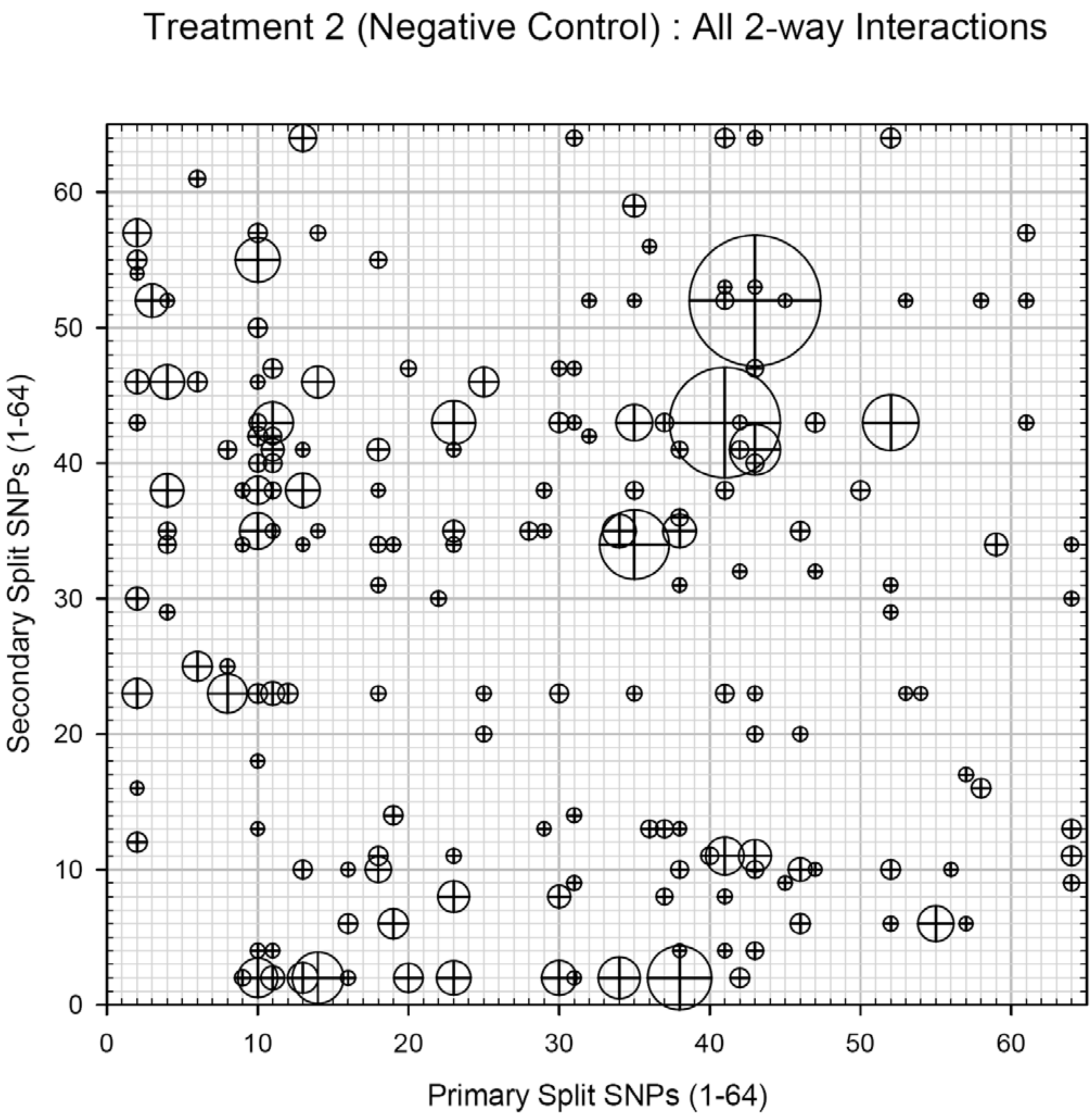
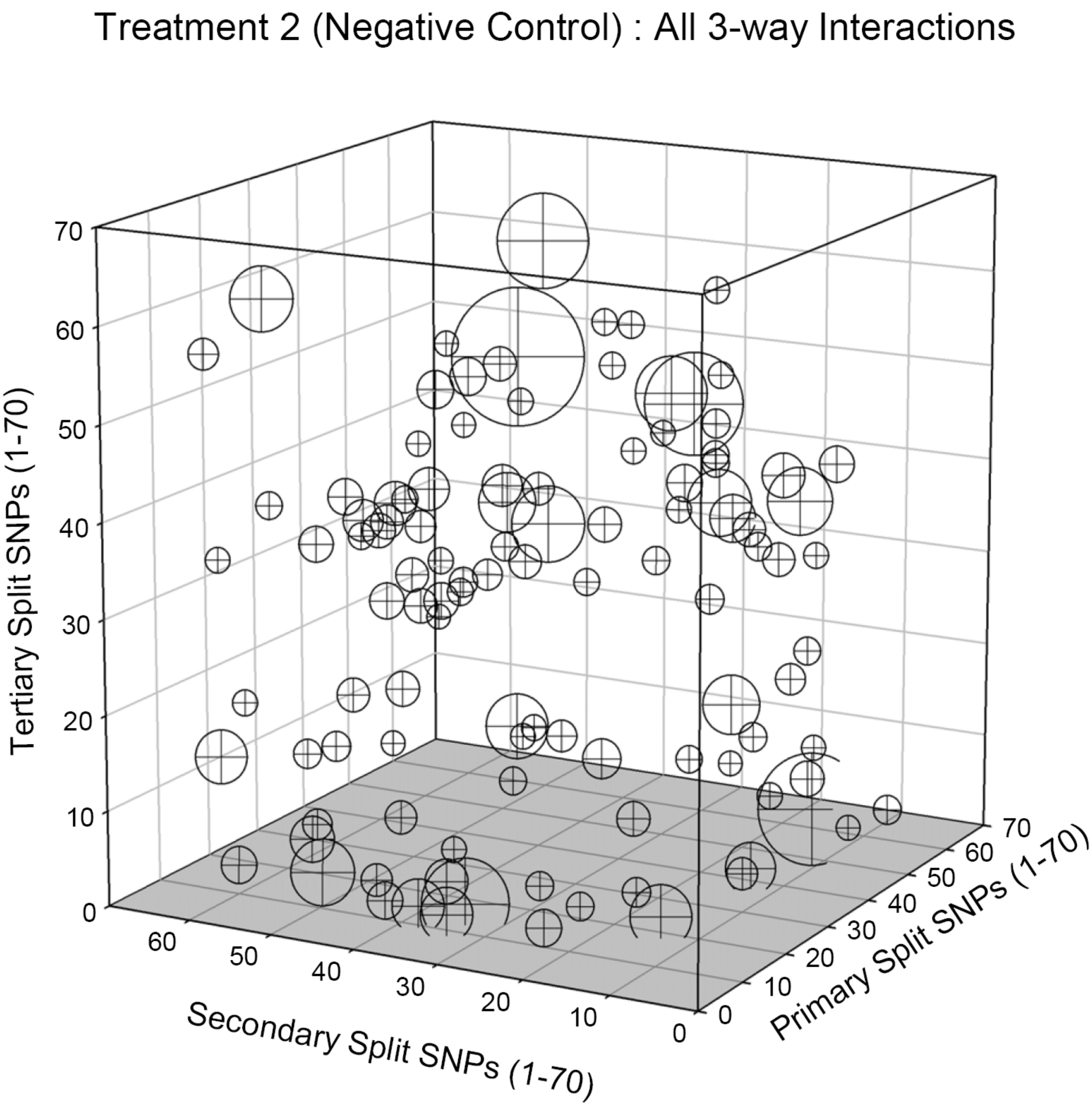

Treatment 2 : All 3-way Interactions
Tertiary Split SNPs (1-70)
Secondary Split SNPs (1-70)
Primary Split SNPs (1-70)
0
10
20
30
40
50
60
70

Figure 9.



Treatment 2 (Negative Control) : All 2-way Interactions
Secondary Split SNPs (1-64)
Primary Split SNPs (1-64)
0
10
20
30
40
50
60

Figure 10.



Treatment 2 (Negative Control) : All 3-way Interactions
Tertiary Split SNPs (1-70)
70
60
50
40
30
20
10
0
60
50
40
30
20
10
0
Secondary Split SNPs (1-70)
0
10
20
30
40
50
60
70
Primary Split SNPs (1-70)

Figure 11.

Table 3: Correlation of Class Probability Scores with Observed Response, Phenotype and Genotype

	Treatment 1	Treatment 2	Treatment 2 (Control)
Observed Response			
Random Forest	0.141	0.231	0.082
Stepwise Logistic	0.148	0.205	0.061
RP Algorithm	0.31	0.325	0.12
True Responder Class			
Random Forest	0.36	0.39	0.094
Stepwise Logistic	0.273	0.373	0.077
RP Algorithm	0.445	0.557	0.086
Genotype Prob (Responder Class)			
Random Forest	0.45	0.608	0.044
Stepwise Logistic	0.26	0.618	0.047
RP Algorithm	0.554	0.873	0.075

Overall Distribution of Class Probability Scores by Algorithm and Simulated Treatment

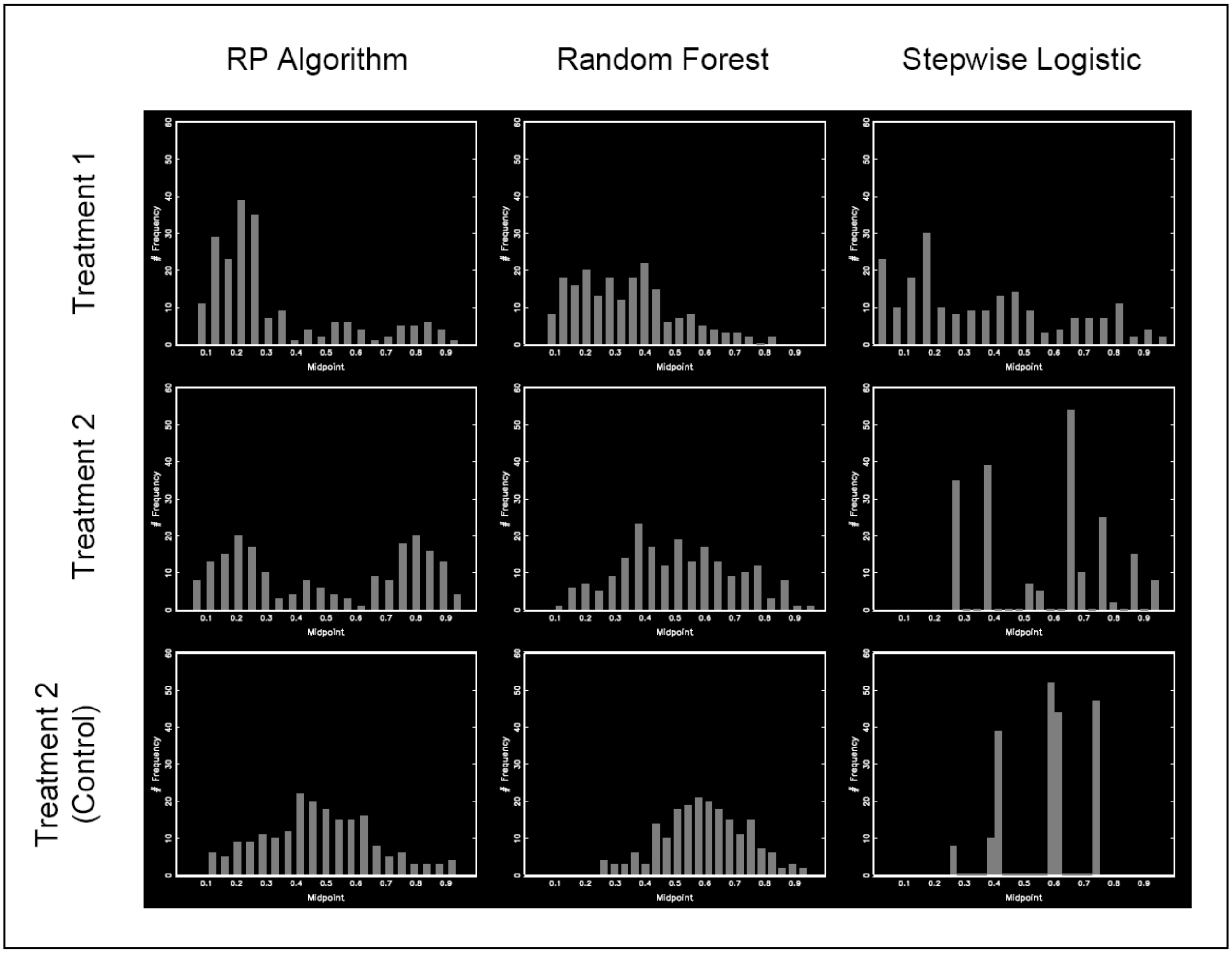


Figure 12

Distribution of Class Probability Scores by Algorithm and Genotype Class Probability for Treatment 1

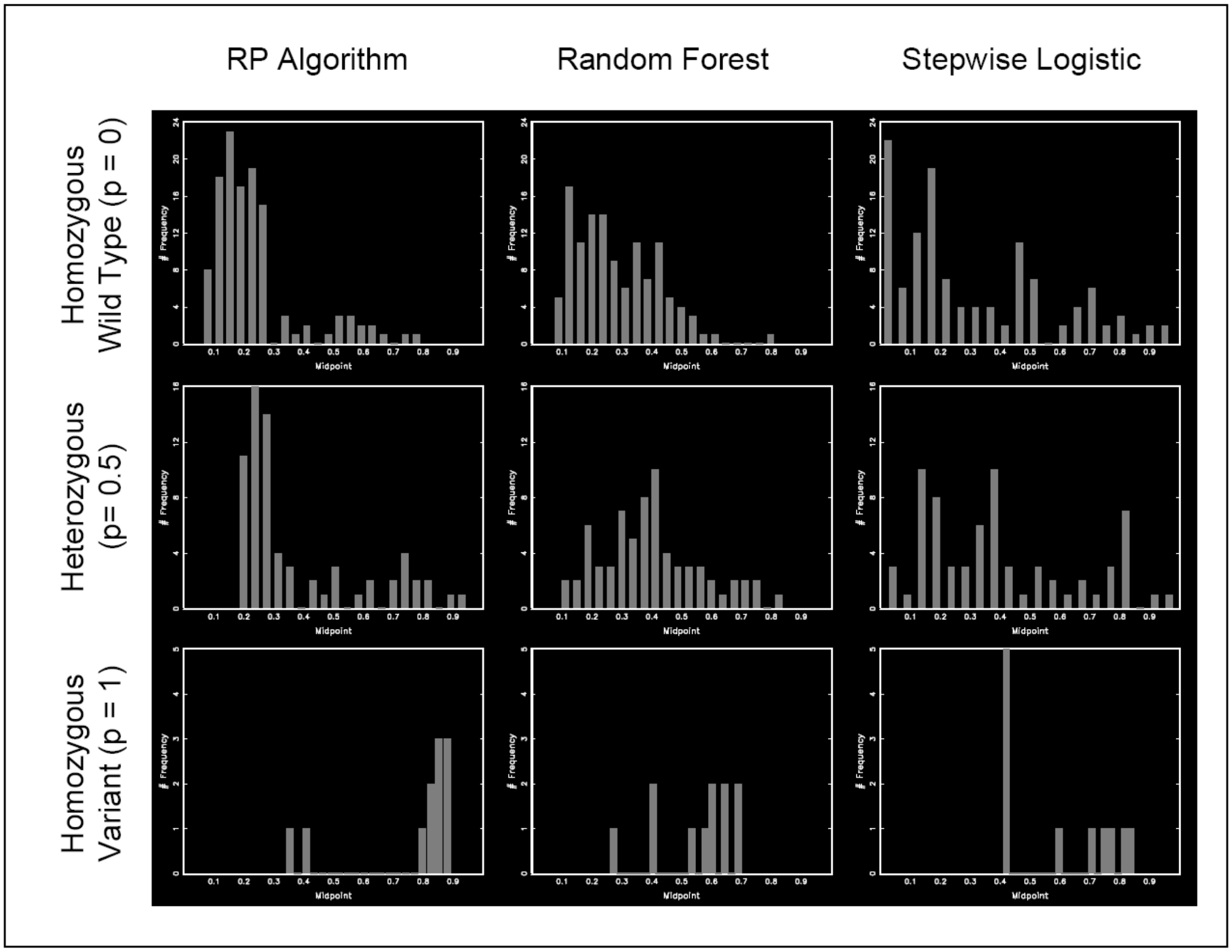


Figure 13

Distribution of Class Probability Scores by Algorithm and Genotype Class Probability for Treatment 2

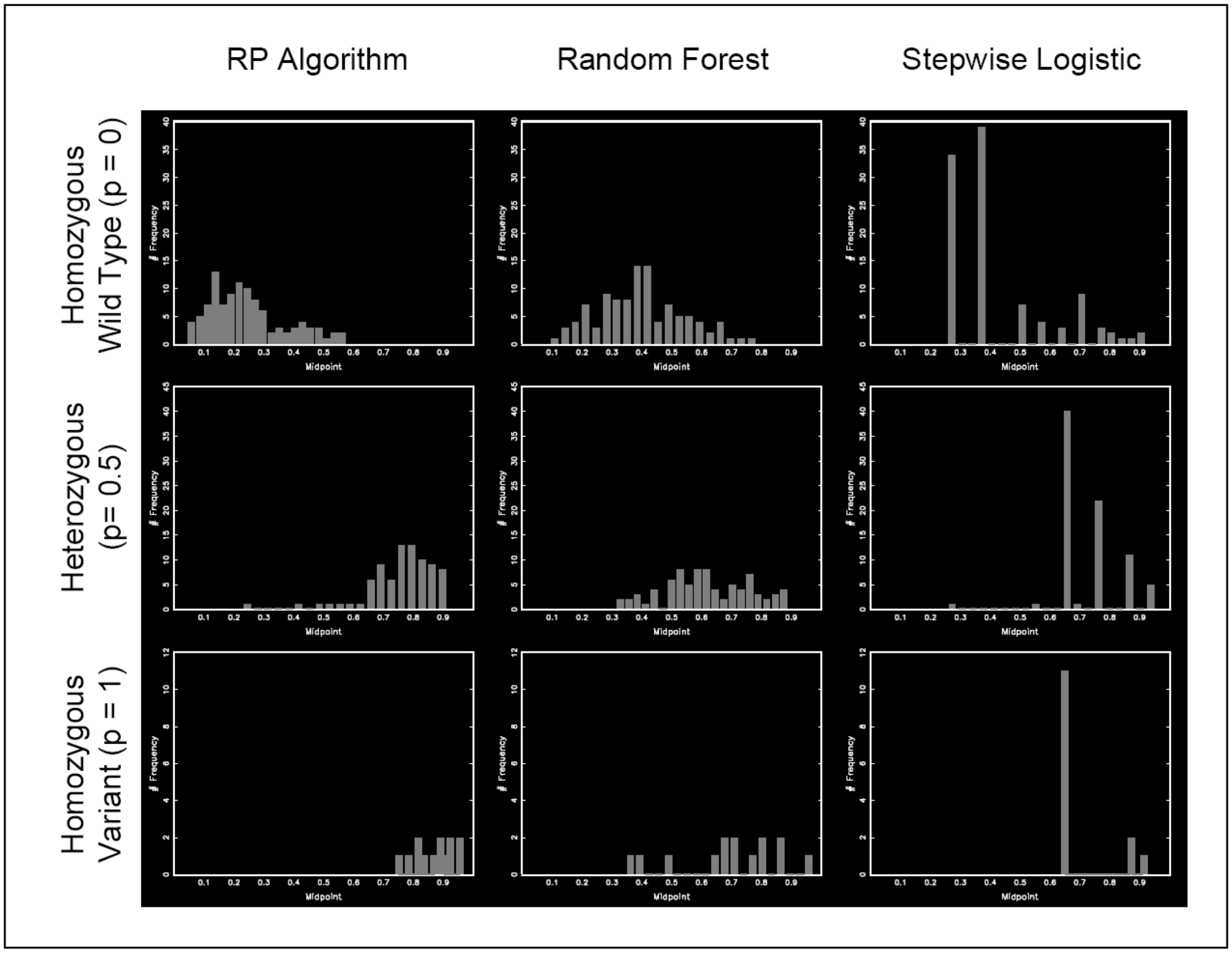


Figure 14

Distribution of Class Probability Scores by Algorithm and Genotype Class Probability for Treatment 2 (Control)

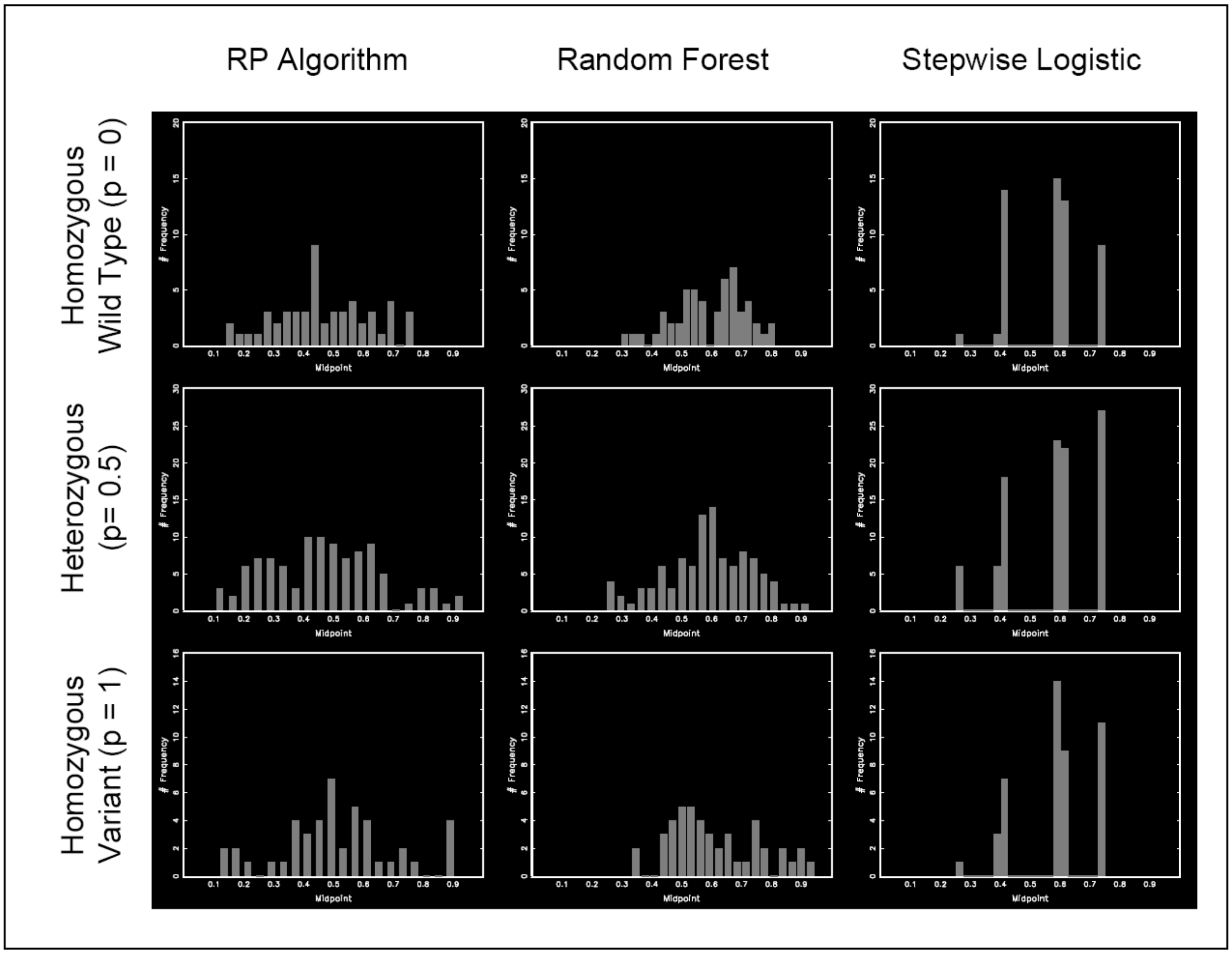


Figure 15

Table 4. MCMC GLM results.

	Posterior Mean	Posterior Std. Dev	Posterior 95% Credible Interval	Monte Carlo Error
Cost Model				
Constant (Treat=1)	-0.025	0.032	[-0.087 , 0.037]	0.001
Effect of Response (Treat=1)	-0.666	0.067	[-0.793 , -0.531]	0.002
Constant (Treat=2)	0.708	0.040	[0.634 , 0.789]	0.001
Effect of Response (Treat=2)	-0.297	0.065	[-0.424 , -0.172]	0.002
Shape Parameter (Treat=1)	6.113	0.603	[4.992 , 7.358]	0.007
Shape Parameter (Treat=2)	5.039	0.490	[4.120 , 6.047]	0.006
QALY Model				
Constant (Treat=1)	-2.204	0.037	[-2.280 , -2.132]	0.002
Effect of Response (Treat=1)	2.589	0.046	[2.497 , 2.678]	0.002
Constant (Treat=2)	-2.165	0.050	[-2.265 , -2.071]	0.003
Effect of Response (Treat=2)	2.597	0.058	[2.489 , 2.711]	0.003
Beta Parameter (Treat=1)	46.890	4.672	[38.270 , 56.590]	0.070
Beta Parameter (Treat=2)	28.960	2.894	[23.570 , 34.950]	0.052
Response Models				
True Class Known				
Constant	-1.314	0.138	[-1.583 , -1.034]	0.007
T (=1 for Treatment 2)	0.486	0.181	[0.117 , 0.847]	0.009
(1-T)*Class (Treat=1)	1.749	0.219	[1.337 , 2.191]	0.009
T*Class (Treat=2)	1.588	0.235	[1.132 , 2.056]	0.007
P(Class | RP Algorithm)				
Constant	-1.331	0.175	[-1.656 , -0.994]	0.011
T (=1 for Treatment 2)	0.155	0.272	[-0.356 , 0.714]	0.018
(1-T)*P(Class | Treat=1)	1.829	0.420	[1.025 , 2.637]	0.024
T*P(Class | Treat=2)	1.490	0.335	[0.817 , 2.134]	0.019
P(Class | Random Forest)				
Constant	-1.087	0.228	[-1.575 , -0.655]	0.018
T (=1 for Treatment 2)	-0.116	0.361	[-0.824 , 0.562]	0.029
(1-T)*P(Class | Treat=1)	1.091	0.583	[0.031 , 2.341]	0.043
T*P(Class | Treat=2)	1.553	0.508	[0.609 , 2.542]	0.036
P(Class | Stepwise Logistic)				
Constant	-0.994	0.168	[-1.341 , -0.663]	0.011
T (=1 for Treatment 2)	-0.187	0.313	[-0.803 , 0.389]	0.024
(1-T)*P(Class | Treat=1)	0.767	0.369	[0.041 , 1.525]	0.022
T*P(Class | Treat=2)	1.353	0.440	[0.515 , 2.202]	0.031

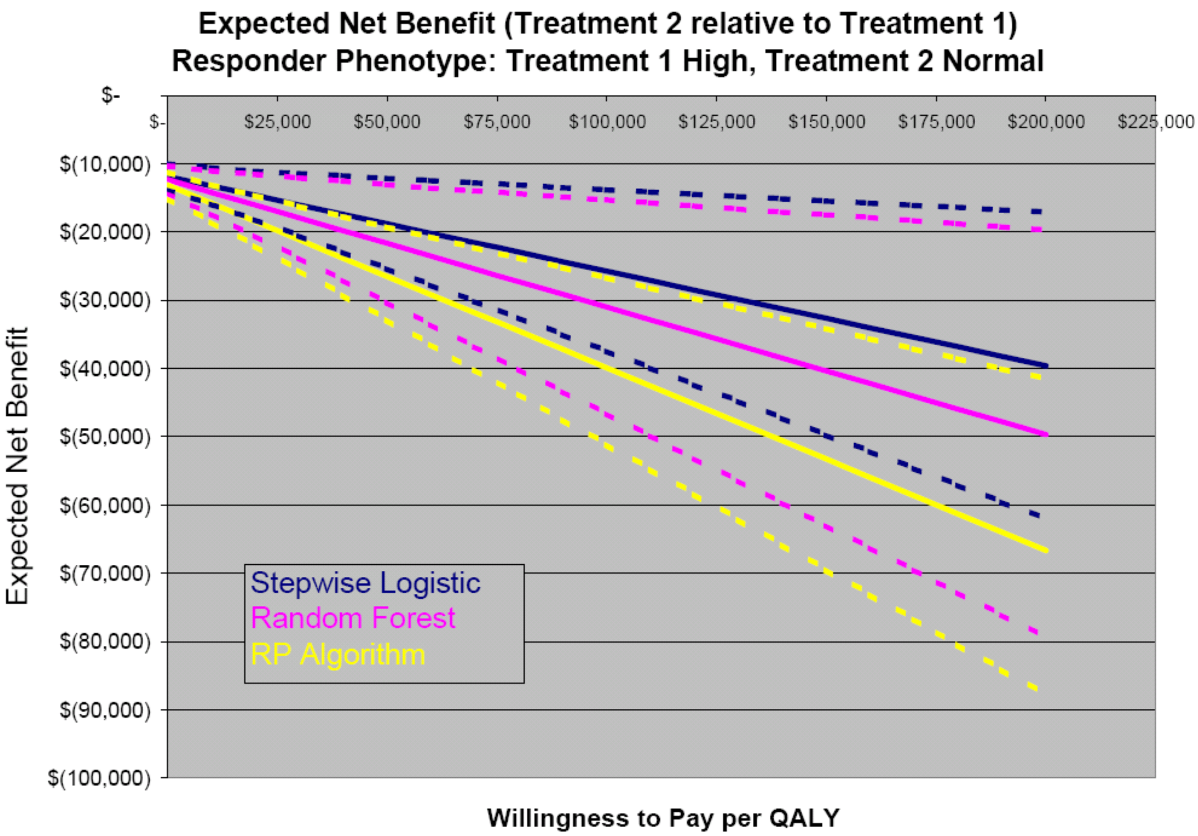
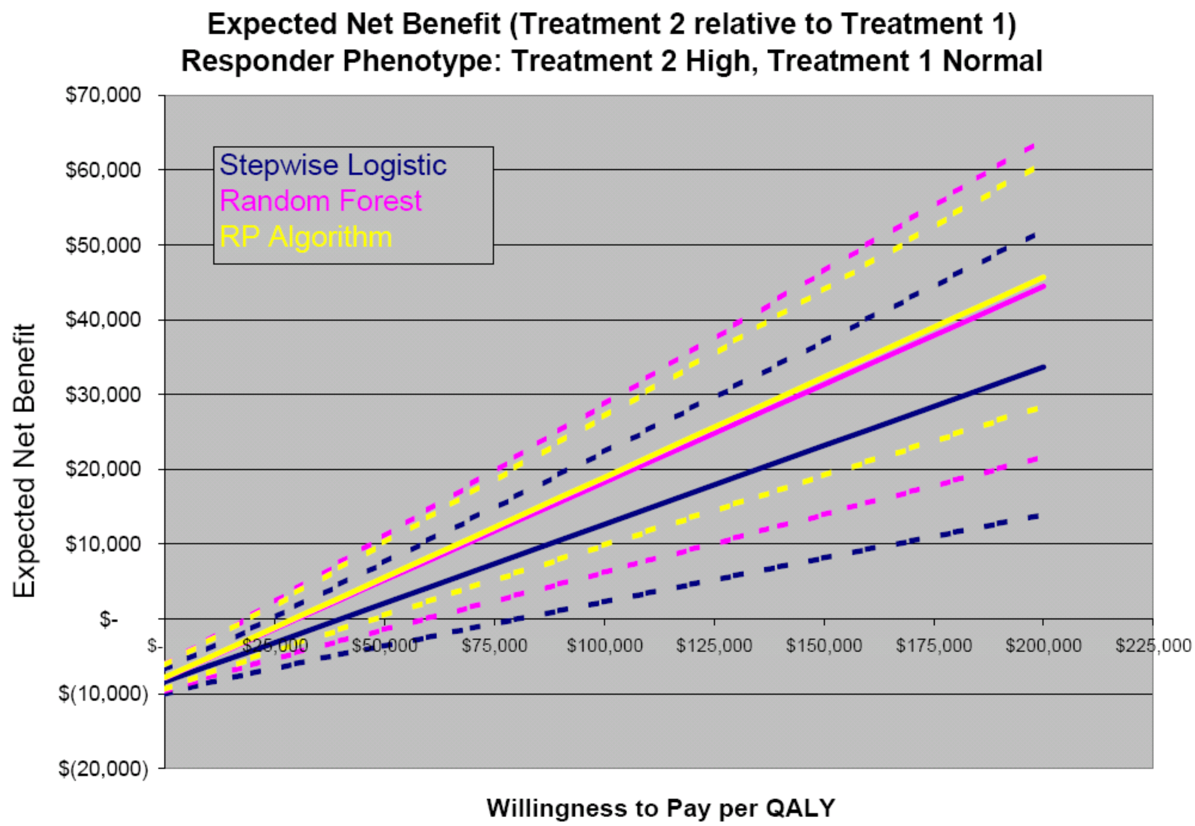

Expected Net Benefit (Treatment 2 relative to Treatment 1)
Responder Phenotype: Treatment 1 High, Treatment 2 Normal
Expected Net Benefit
Willingness to Pay per QALY
$-
$(10,000)
$(20,000)
$(30,000)
$(40,000)
$(50,000)
$(60,000)
$(70,000)
$(80,000)
$(90,000)
$(100,000)
$25,000
$50,000
$75,000
$100,000
$125,000
$150,000
$175,000
$200,000
$225,000
Stepwise Logistic
Random Forest
RP Algorithm

Figure 16



Expected Net Benefit (Treatment 2 relative to Treatment 1)
Responder Phenotype: Treatment 2 High, Treatment 1 Normal
Stepwise Logistic
Random Forest
RP Algorithm
Expected Net Benefit
$70,000
$60,000
$50,000
$40,000
$30,000
$20,000
$10,000
$-
$(10,000)
$(20,000)
$-
$25,000
$50,000
$75,000
$100,000
$125,000
$150,000
$175,000
$200,000
$225,000
Willingness to Pay per QALY

Figure 17

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